

Complete Puppet Mastery Part – 2

Practical 9 - Monitoring and Managing Apache Web Server Using Puppet

```
ubuntu@ip-172-31-2-124: ~  
ubuntu@ip-172-31-2-124:~$ cd /etc/puppetlabs/code/environments/production/modules  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ sudo  
mkdir webserver  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules$ cd  
webserver  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver$ sudo mkdir manifests templates files  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver$ ls  
files manifests templates  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver$ tree  
.  
├── files  
├── manifests  
└── templates  
  
4 directories, 0 files  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver$ cd manifests  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver/manifests$ sudo touch init.pp  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse
```

```
ubuntu@ip-172-31-2-124: ~  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver/manifests$ sudo touch init.pp  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver/manifests$ sudo touch install.pp  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver/manifests$ sudo touch config.pp  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver/manifests$ sudo touch service.pp  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver/manifests$ ls  
config.pp init.pp install.pp service.pp  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver/manifests$ sudo nano install.pp  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver/manifests$ cd templates  
-bash: cd: templates: No such file or directory  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver/manifests$ cd ..  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver$ cd templates  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver/templates$ sudo nano index.html.epp  
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webse  
rver/templates$ cd ..
```

```
ubuntu@ip-172-31-2-124: ~
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webserver/manifests$ sudo /opt/puppetlabs/bin/puppet parser validate init.pp
Error: Could not parse for environment production: The parameter $title redefines a built in parameter in the Host Class Definition (file: /etc/puppetlabs/code/environments/production/modules/webserver/manifests/init.pp, line: 2, column: 10)
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webserver/manifests$ sudo nano init.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webserver/manifests$ sudo nano init.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webserver/manifests$ sudo /opt/puppetlabs/bin/puppet parser validate init.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webserver/manifests$ sudo nano config.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webserver/manifests$ sudo /opt/puppetlabs/bin/puppet parser validate config.pp
Error: Could not parse for environment production: Syntax error at end of input (file: /etc/puppetlabs/code/environments/production/modules/webserver/manifests/config.pp)
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webserver/manifests$ sudo nano config.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webserver/manifests$ sudo /opt/puppetlabs/bin/puppet parser validate config.pp
ubuntu@ip-172-31-2-124:/etc/puppetlabs/code/environments/production/modules/webserver/manifests$
```

```
ubuntu@ip-172-31-2-124: ~
ubuntu@ip-172-31-2-124:~$ sudo nano site.pp
ubuntu@ip-172-31-2-124:~$ puppet parser validate site.pp
ubuntu@ip-172-31-2-124:~$ sudo /opt/puppetlabs/bin/puppet apply site.pp
Notice: Compiled catalog for ip-172-31-2-124.ap-south-1.compute.internal in environment production in 0.47 seconds
Notice: /Stage[main]/Webserver::Install/Package[apache2]/ensure: created
Notice: /Stage[main]/Webserver::Config/File[/var/www/html/index.html]/content: content changed '{sha256}6faef4d5d777fdcaa653766b0ac8b9ed32d0fd87f7dcd79f02ff524dd1b0eb69' to '{sha256}61684a401d9064330f06c19e278de7993c5de855ebcd0e896771005ac862dcbe'
Notice: /Stage[main]/Webserver::Service/Service[apache2]: Triggered 'refresh' from 1 event
Notice: Applied catalog in 14.22 seconds
ubuntu@ip-172-31-2-124:~$ systemctl status apach2
Unit apach2.service could not be found.
ubuntu@ip-172-31-2-124:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: >
   Active: active (running) since Sat 2026-08-01 11:14:52 UTC; 36s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 2250 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SU>
  Main PID: 2254 (apache2)
    Tasks: 55 (limit: 627)
   Memory: 5.4M (peak: 5.6M)
```



```
ubuntu@ip-172-31-2-124: ~  
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/code/environments/production/  
modules/webserver/manifests/init.pp  
class webserver(  
  String $site_title="Vishnu's puppet web server",  
  String $heading="Welcome!",  
  String $environment="Development",  
  String $hostname="localhost"  
) {  
  include webserver::install  
  
  class {'webserver::config':  
    site_title=>$site_title,  
    heading=>$heading,  
    environment=>$environment,  
    hostname=>$hostname  
  }  
  
  include webserver::service  
}  
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/code/environments/production/  
modules/webserver/manifests/config.pp  
class webserver::config(  
  String $site_title,  
  String $heading,
```

```
ubuntu@ip-172-31-2-124: ~  
}  
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/code/environments/production/  
modules/webserver/manifests/config.pp  
class webserver::config(  
  String $site_title,  
  String $heading,  
  String $environment,  
  String $hostname  
) {  
  file{'/var/www/html/index.html':  
    ensure=>file,  
    owner=>'root',  
    group=>'root',  
    mode=>'0644',  
    content=>epp(  
      'webserver/index.html.epp',  
      {  
        'site_title'=>$site_title,  
        'heading'=>$heading,  
        'environment'=>$environment,  
        'hostname'=>$hostname  
      }  
    ),  
    require=>Class['webserver::install'],  
  }  
}
```

```
ubuntu@ip-172-31-2-124: ~  
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/code/environments/production/  
modules/webserver/manifests/install.pp  
class webserver::install {  
  package{'apache2':  
    ensure=>installed  
  }  
}  
ubuntu@ip-172-31-2-124:~$ sudo cat /etc/puppetlabs/code/environments/production/  
modules/webserver/manifests/service.pp  
class webserver::service {  
  service{'apache2':  
    ensure=>running,  
    enable=>true,  
    require=>Class['webserver::install']  
  }  
}  
ubuntu@ip-172-31-2-124:~$ sudo cat site.pp  
class{'webserver':  
  site_title=>"Production Apache server",  
  heading=>"Welcome to puppet managed infra",  
  environment=>"Production",  
  hostname=>"web01"  
}  
ubuntu@ip-172-31-2-124:~$
```

Practical-10: Deploying and Configuring a Secure Puppet Master-Agent Infrastructure

Although the tasks in this practical were performed in parallel on the Puppet Master and Puppet Agent, I have organized the documentation by presenting all the Puppet Master screenshots first, followed by the Puppet Agent screenshots, for better readability, also this practical would stay so close to my heart as I literally learnt so a lot after a lot of debugging.

Puppetserver:

```
ubuntu@ip-172-31-14-115: ~
sreev@admin MINGW64 ~ (main)
$ cd d:

sreev@admin MINGW64 /d
$ cd downloads

sreev@admin MINGW64 /d/downloads
$ ssh -i "puppet-lab-key.pem" ubuntu@ec2-3-110-188-5.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-3-110-188-5.ap-south-1.compute.amazonaws.com (3.110.188.5)' can't be established.
ED25519 key fingerprint is SHA256:ouWrQ8ItMaigOVEhLqZeoywJlkHUYh/ZGtoWfidkZBI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-110-188-5.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1061-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Aug  3 16:03:24 UTC 2026

System load:  0.45          Processes:            115
Usage of /:   22.9% of 7.57GB Users logged in:        0
Memory usage: 12%          IPv4 address for ens5: 172.31.14.115
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
```



```
ubuntu@ip-172-31-14-115: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
ubuntu@ip-172-31-14-115:~$ hostname  
ip-172-31-14-115  
ubuntu@ip-172-31-14-115:~$ hostname -f  
ip-172-31-14-115.ap-south-1.compute.internal  
ubuntu@ip-172-31-14-115:~$ sudo hostnamectl set-hostname puppet-maste  
ubuntu@ip-172-31-14-115:~$ sudo hostnamectl set-hostname puppet-master  
ubuntu@ip-172-31-14-115:~$ hostname  
puppet-master  
ubuntu@ip-172-31-14-115:~$ hostname -f  
puppet-master  
ubuntu@ip-172-31-14-115:~$ sudo nano /etc/hosts  
ubuntu@ip-172-31-14-115:~$ wget https://apt.puppet.com/puppet8-release-jammy.deb  
--2026-08-03 16:07:18-- https://apt.puppet.com/puppet8-release-jammy.deb  
Resolving apt.puppet.com (apt.puppet.com)... 13.225.5.12, 13.225.5.15, 13.225.5.  
20, ...  
Connecting to apt.puppet.com (apt.puppet.com)|13.225.5.12|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 3786 (3.7K) [application/vnd.debian.binary-package]  
Saving to: 'puppet8-release-jammy.deb'  
  
puppet8-release-jam 100%[=====>] 3.70K --.-KB/s in 0s  
2026-08-03 16:07:18 (1.89 GB/s) - 'puppet8-release-jammy.deb' saved [3786/3786]  
  
ubuntu@ip-172-31-14-115:~$ sudo dpkg -i puppet8-release-jammy.deb  
Selecting previously unselected package puppet8-release.  
(Reading database ... 66021 files and directories currently installed.)  
Preparing to unpack puppet8-release-jammy.deb ...  
Unpacking puppet8-release (1.0.0-10jammy) ...
```

```
ubuntu@ip-172-31-14-115: ~  
Get:37 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 c-n-f M  
etadata [544 B]  
Fetched 35.3 MB in 6s (6092 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
All packages are up to date.  
ubuntu@ip-172-31-14-115:~$ sudo apt install puppetserver -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
alsa-topology-conf alsa-ucm-conf ca-certificates-java fontconfig-config  
fonts-dejavu-core java-common libasound2 libasound2-data libavahi-client3  
libavahi-common-data libavahi-common3 libcups2 libfontconfig1 libgraphite2-3  
libharfbuzz0b libjpeg-turbo8 libjpeg8 liblcms2-2 libpcsclite1 net-tools  
openjdk-17-jre-headless puppet-agent  
Suggested packages:  
default-jre libasound2-plugins alsa-utils cups-common liblcms2-utils pcsd  
libnss-mdns fonts-dejavu-extra fonts-ipafont-gothic fonts-ipafont-mincho  
fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic  
The following NEW packages will be installed:  
alsa-topology-conf alsa-ucm-conf ca-certificates-java fontconfig-config  
fonts-dejavu-core java-common libasound2 libasound2-data libavahi-client3  
libavahi-common-data libavahi-common3 libcups2 libfontconfig1 libgraphite2-3  
libharfbuzz0b libjpeg-turbo8 libjpeg8 liblcms2-2 libpcsclite1 net-tools  
openjdk-17-jre-headless puppet-agent puppetserver  
0 upgraded, 23 newly installed, 0 to remove and 0 not upgraded.  
Need to get 164 MB of archives.  
After this operation, 429 MB of additional disk space will be used.  
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 alsa-topo  
logy-conf all 1.2.5.1-2 [15.5 kB]
```

```
ubuntu@ip-172-31-14-115: ~  
No user sessions are running outdated binaries.  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-14-115:~$ sudo nano /opt/puppetlabs/puppet/puppet.conf  
ubuntu@ip-172-31-14-115:~$ sudo nano /etc/puppetlabs/puppet/puppet.conf  
ubuntu@ip-172-31-14-115:~$ sudo /opt/puppetlabs/bin/puppet config print certname  
puppet-master  
ubuntu@ip-172-31-14-115:~$ sudo nano /etc/default/puppetserver  
ubuntu@ip-172-31-14-115:~$ sudo systemctl start puppetserver  
ubuntu@ip-172-31-14-115:~$ sudo systemctl status puppetserver  
● puppetserver.service - puppetserver Service  
   Loaded: loaded (/lib/systemd/system/puppetserver.service; disabled; vendor  
   Active: active (running) since Mon 2026-08-03 16:13:31 UTC; 12s ago  
   Process: 3632 ExecStart=/opt/puppetlabs/server/apps/puppetserver/bin/puppet  
   Main PID: 3693 (java)  
    Tasks: 62 (limit: 4915)  
   Memory: 845.9M  
    CPU: 1min 3.920s  
   CGroup: /system.slice/puppetserver.service  
           └─3693 /usr/bin/java --add-opens java.base/sun.nio.ch=ALL-UNNAMED >  
             └─3828 /opt/puppetlabs/puppet/bin/ruby /opt/puppetlabs/server/data>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,543 |-INFO in ch.qos>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,543 |-INFO in ch.qos>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,545 |-INFO in ch.qos>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,546 |-INFO in ch.qos>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,552 |-INFO in ch.qos>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,552 |-INFO in ch.qos>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,552 |-INFO in ch.qos>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,552 |-INFO in ch.qos>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,552 |-INFO in ch.qos>
```

```
ubuntu@ip-172-31-14-115: ~  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,552 |-INFO in ch.qos>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,552 |-INFO in ch.qos>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,552 |-INFO in ch.qos>  
Aug 03 16:13:30 puppet-master puppetserver[3693]: 16:13:30,552 |-INFO in ch.qos>  
Aug 03 16:13:31 puppet-master systemd[1]: Started puppetserver Service.  
ubuntu@ip-172-31-14-115:~$ sudo ss -tlnp | grep 8140  
LISTEN 0      50          *:8140      *:~      users:(("java",pid=369  
3,fd=27))  
ubuntu@ip-172-31-14-115:~$ sudo ls -R /etc/puppetlabs/puppet/ssl  
/etc/puppetlabs/puppet/ssl:  
ca certificate_requests certs crl.pem private private_keys public_keys  
  
/etc/puppetlabs/puppet/ssl/certificate_requests:  
  
/etc/puppetlabs/puppet/ssl/certs:  
ca.pem puppet-master.pem  
  
/etc/puppetlabs/puppet/ssl/private:  
  
/etc/puppetlabs/puppet/ssl/private_keys:  
puppet-master.pem  
  
/etc/puppetlabs/puppet/ssl/public_keys:  
puppet-master.pem  
ubuntu@ip-172-31-14-115:~$ sudo /opt/puppetlabs/bin/puppetserver ca list  
Requested Certificates:  
  puppet-agent01 (SHA256) F1:16:91:90:DA:8C:E5:DE:A5:97:D7:C3:FE:D2:37:  
66:4C:B4:57:E5:B7:67:36:DF:D2:8C:A1:F8:2F:FD:C9:E3  
ubuntu@ip-172-31-14-115:~$ sudo /opt/puppetlabs/bin/puppetserver ca sign --certn  
ame puppet-agent01  
Successfully signed certificate request for puppet-agent01  
ubuntu@ip-172-31-14-115:~$ |
```

```
ubuntu@ip-172-31-14-115: ~  
ubuntu@ip-172-31-14-115:~$ sudo /opt/puppetlabs/bin/puppetserver ca sign --certname puppet-agent01  
Successfully signed certificate request for puppet-agent01  
ubuntu@ip-172-31-14-115:~$ sudo cat /etc/puppetlabs/puppet/puppet.conf  
# This file can be used to override the default puppet settings.  
# See the following links for more details on what settings are available:  
# - https://puppet.com/docs/puppet/latest/config_important_settings.html  
# - https://puppet.com/docs/puppet/latest/config_about_settings.html  
# - https://puppet.com/docs/puppet/latest/config_file_main.html  
# - https://puppet.com/docs/puppet/latest/configuration.html  
[main]  
certname = puppet-master  
server = puppet-master  
[server]  
vardir = /opt/puppetlabs/server/data/puppetserver  
logdir = /var/log/puppetlabs/puppetserver  
rundir = /var/run/puppetlabs/puppetserver  
pidfile = /var/run/puppetlabs/puppetserver/puppetserver.pid  
codedir = /etc/puppetlabs/code  
ubuntu@ip-172-31-14-115:~$ sudo cat /etc/hosts  
127.0.0.1 localhost  
  
# The following lines are desirable for IPv6 capable hosts  
::1 ip6-localhost ip6-loopback  
fe00::0 ip6-localnet  
ff00::0 ip6-mcastprefix  
ff02::1 ip6-allnodes  
ff02::2 ip6-allrouters  
ff02::3 ip6-allhosts  
172.31.14.115 puppet-master  
172.31.7.84 puppet-agent01  
ubuntu@ip-172-31-14-115:~$ |
```

Agent:

```
ubuntu@ip-172-31-7-84: ~  
sreev@admin MINGW64 ~ (main)  
$ cd d:  
  
sreev@admin MINGW64 /d  
$ cd downloads  
  
sreev@admin MINGW64 /d/downloads  
$ ssh -i "puppet-lab-key.pem" ubuntu@ec2-13-127-65-166.ap-south-1.compute.amazonaws.com  
The authenticity of host 'ec2-13-127-65-166.ap-south-1.compute.amazonaws.com (13.127.65.166)' can't be established.  
ED25519 key fingerprint is SHA256:uhqvsB1y8DUB+VL1DVyNU5q6W51PFpOC1wsI3bmR1RI.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'ec2-13-127-65-166.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1061-aws x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Mon Aug  3 15:15:11 UTC 2026
```



```
ubuntu@ip-172-31-7-84: ~
ubuntu@ip-172-31-7-84:~$ uname -r
6.8.0-1061-aws
ubuntu@ip-172-31-7-84:~$ uname -m
x86_64
ubuntu@ip-172-31-7-84:~$ hostnamectl
  Static hostname: ip-172-31-7-84
        Icon name: computer-vm
        Chassis: vm
        Machine ID: ec25a049344db437c891be130636104e
        Boot ID: 397bcef74ffb473394db450132cac429
  Virtualization: amazon
  Operating System: Ubuntu 22.04.5 LTS
        Kernel: Linux 6.8.0-1061-aws
  Architecture: x86-64
  Hardware Vendor: Amazon EC2
  Hardware Model: t3.micro
ubuntu@ip-172-31-7-84:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           914Mi       188Mi       115Mi       2.0Mi       610Mi       566Mi
Swap:              0B           0B           0B
```

```
ubuntu@ip-172-31-7-84: ~
tmpfs          92M 4.0K 92M 1% /run/user/1000
ubuntu@ip-172-31-7-84:~$ hostname
ip-172-31-7-84
ubuntu@ip-172-31-7-84:~$ hostname -f
ip-172-31-7-84.ap-south-1.compute.internal
ubuntu@ip-172-31-7-84:~$ hostname -I
172.31.7.84
ubuntu@ip-172-31-7-84:~$ sudo hostnamectl set-hostname puppet-agent01
ubuntu@ip-172-31-7-84:~$ hostname
puppet-agent01
ubuntu@ip-172-31-7-84:~$ sudo nano /etc/hosts
ubuntu@ip-172-31-7-84:~$ sudo apt install puppet-agent -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package puppet-agent
ubuntu@ip-172-31-7-84:~$ wget https://apt.puppet.com/puppet8-release-jammy.deb
--2026-08-03 15:35:28-- https://apt.puppet.com/puppet8-release-jammy.deb
Resolving apt.puppet.com (apt.puppet.com)... 13.225.5.127, 13.225.5.15, 13.225.5.12, ...
Connecting to apt.puppet.com (apt.puppet.com)|13.225.5.127|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3786 (3.7K) [application/vnd.debian.binary-package]
Saving to: 'puppet8-release-jammy.deb'
```

```
ubuntu@ip-172-31-7-84: ~  
directory  
ubuntu@ip-172-31-7-84:~$ sudo dpkg -i puppet8-release-jammy.deb  
Selecting previously unselected package puppet8-release.  
(Reading database ... 66021 files and directories currently installed.)  
Preparing to unpack puppet8-release-jammy.deb ...  
Unpacking puppet8-release (1.0.0-10jammy) ...  
Setting up puppet8-release (1.0.0-10jammy) ...  
ubuntu@ip-172-31-7-84:~$ sudo apt update  
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease  
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]  
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]  
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]  
Get:5 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]  
Get:6 http://apt.puppet.com jammy InRelease [113 kB]  
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]  
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]  
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]  
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [5652 kB]
```

```
ubuntu@ip-172-31-7-84: ~  
All packages are up to date.  
ubuntu@ip-172-31-7-84:~$ sudo apt install puppet-agent -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following NEW packages will be installed:  
  puppet-agent  
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.  
Need to get 36.0 MB of archives.  
After this operation, 125 MB of additional disk space will be used.  
Get:1 http://apt.puppet.com jammy/puppet8 amd64 puppet-agent amd64 8.10.0-1jammy [36.0 MB]  
Fetched 36.0 MB in 0s (89.4 MB/s)  
Selecting previously unselected package puppet-agent.  
(Reading database ... 66026 files and directories currently installed.)  
Preparing to unpack .../puppet-agent_8.10.0-1jammy_amd64.deb ...  
Unpacking puppet-agent (8.10.0-1jammy) ...  
Setting up puppet-agent (8.10.0-1jammy) ...  
Created symlink /etc/systemd/system/multi-user.target.wants/pxp-agent.service → /lib/systemd/system/pxp-agent.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/puppet.service → /lib/systemd/system/puppet.service.  
Removed /etc/systemd/system/multi-user.target.wants/pxp-agent.service.  
Processing triggers for libc-bin (2.35-0ubuntu3.14) ...
```

```
ubuntu@ip-172-31-7-84: ~  
ubuntu@ip-172-31-7-84:~$ sudo nano /etc/hosts  
ubuntu@ip-172-31-7-84:~$ sudo /etc/default/puppetserver  
[sudo] password for ubuntu:  
sudo: a password is required  
ubuntu@ip-172-31-7-84:~$ sudo cat /etc/default/puppetserver  
cat: /etc/default/puppetserver: No such file or directory  
ubuntu@ip-172-31-7-84:~$ sudo nano /etc/puppetlabs/puppet/puppet.conf  
ubuntu@ip-172-31-7-84:~$ sudo /opt/puppetlabs/bin/puppet config print certname  
puppet-agent01  
ubuntu@ip-172-31-7-84:~$ sudo /opt/puppetlabs/bin/puppet config print server  
puppet-master  
ubuntu@ip-172-31-7-84:~$ getent hosts puppet-master  
172.31.14.115 puppet-master  
ubuntu@ip-172-31-7-84:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Refreshed CRL: 44:14:C7:B4:D2:C6:A9:A1:58:64:DD:CD:B7:C5:B6:8B:BB:B8:7A:20  
:D4:63:9A:A7:2E:6B:2B:BF:11:98:9F:A5  
Info: Creating a new RSA SSL key for puppet-agent01  
Info: csr_attributes file loading from /etc/puppetlabs/puppet/csr_attributes.yaml  
Info: Creating a new SSL certificate request for puppet-agent01  
Info: Certificate Request fingerprint (SHA256): F1:16:91:90:DA:8C:E5:DE:A5:97:D7  
:C3:FE:D2:37:66:4C:B4:57:E5:B7:67:36:DF:D2:8C:A1:F8:2F:FD:C9:E3  
Info: Certificate for puppet-agent01 has not been signed yet  
Couldn't fetch certificate from CA server; you might still need to sign this age
```

Finally:

```
Info: Certificate Request fingerprint (SHA256): F1:16:91:90:DA:8C:E5:DE:A5:97:D7  
:C3:FE:D2:37:66:4C:B4:57:E5:B7:67:36:DF:D2:8C:A1:F8:2F:FD:C9:E3  
Info: Certificate for puppet-agent01 has not been signed yet  
Couldn't fetch certificate from CA server; you might still need to sign this age  
nt's certificate (puppet-agent01).  
Exiting now because the waitforcert setting is set to 0.  
ubuntu@ip-172-31-7-84:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: csr_attributes file loading from /etc/puppetlabs/puppet/csr_attributes.yaml  
Info: Creating a new SSL certificate request for puppet-agent01  
Info: Certificate Request fingerprint (SHA256): F1:16:91:90:DA:8C:E5:DE:A5:97:D7  
:C3:FE:D2:37:66:4C:B4:57:E5:B7:67:36:DF:D2:8C:A1:F8:2F:FD:C9:E3  
Info: Downloaded certificate for puppet-agent01 from https://puppet-master:8140/  
puppet-ca/v1  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1785774353'  
Info: Creating state file /opt/puppetlabs/puppet/cache/state/state.yaml  
Notice: Applied catalog in 0.01 seconds  
ubuntu@ip-172-31-7-84:~$
```

Practical 11 - Centralized Configuration Management Using Puppet Master Catalogs

Puppetserver:

```
ubuntu@ip-172-31-14-115: ~  
ubuntu@ip-172-31-14-115:~$ sudo /opt/puppetlabs/bin/puppet config print environmentpath  
/etc/puppetlabs/code/environments  
ubuntu@ip-172-31-14-115:~$ ls -l /etc/puppetlabs/code/environments  
total 4  
drwxr-xr-x 5 root root 4096 Aug  3 16:08 production  
ubuntu@ip-172-31-14-115:~$ cd /etc/puppetlabs/code/environments/production/modules  
ubuntu@ip-172-31-14-115:/etc/puppetlabs/code/environments/production/modules$ sudo mkdir -p centraldemo/manifests  
ubuntu@ip-172-31-14-115:/etc/puppetlabs/code/environments/production/modules$ tree centraldemo  
Command 'tree' not found, but can be installed with:  
sudo apt install tree  
ubuntu@ip-172-31-14-115:/etc/puppetlabs/code/environments/production/modules$ ls -l  
total 4  
drwxr-xr-x 3 root root 4096 Aug  3 17:12 centraldemo  
ubuntu@ip-172-31-14-115:/etc/puppetlabs/code/environments/production/modules$ find centraldemo  
centraldemo  
centraldemo/manifests  
ubuntu@ip-172-31-14-115:/etc/puppetlabs/code/environments/production/modules$ cd manifests  
-bash: cd: manifests: No such file or directory  
ubuntu@ip-172-31-14-115:/etc/puppetlabs/code/environments/production/modules$ cd centraldemo  
mo  
ubuntu@ip-172-31-14-115:/etc/puppetlabs/code/environments/production/modules/centraldemo$  
cd manifests  
ubuntu@ip-172-31-14-115:/etc/puppetlabs/code/environments/production/modules/centraldemo/manifests$ sudo nano init.pp  
ubuntu@ip-172-31-14-115:/etc/puppetlabs/code/environments/production/modules/centraldemo/manifests$ cd ..  
ubuntu@ip-172-31-14-115:/etc/puppetlabs/code/environments/production/modules/centraldemo$  
cd ..  
ubuntu@ip-172-31-14-115:/etc/puppetlabs/code/environments/production/modules$ mkdir manife
```

```
ubuntu@ip-172-31-14-115: ~  
ubuntu@ip-172-31-14-115:~$ ls -R /etc/puppetlabs/code/environments/production  
/etc/puppetlabs/code/environments/production:  
data environment.conf hiera.yaml manifests modules  
  
/etc/puppetlabs/code/environments/production/data:  
  
/etc/puppetlabs/code/environments/production/manifests:  
site.pp  
  
/etc/puppetlabs/code/environments/production/modules:  
centraldemo  
  
/etc/puppetlabs/code/environments/production/modules/centraldemo:  
manifests  
  
/etc/puppetlabs/code/environments/production/modules/centraldemo/manifests:  
init.pp  
ubuntu@ip-172-31-14-115:~$ sudo nano /etc/puppetlabs/code/environments/production/manifests/site.pp  
ubuntu@ip-172-31-14-115:~$ sudo nano /etc/puppetlabs/code/environments/production/manifests/site.pp  
ubuntu@ip-172-31-14-115:~$ sudo cat /etc/puppetlabs/code/environments/production/manifests/site.pp  
node 'puppet-agent01'{  
  include centraldemo  
}  
ubuntu@ip-172-31-14-115:~$ sudo cat /etc/puppetlabs/code/environments/production/modules/centraldemo/manifests/init.pp  
class centraldemo{  
  package{'tree':  
    ensure=>installed  
  }  
}
```

```
ubuntu@ip-172-31-14-115: ~  
ubuntu@ip-172-31-14-115:~$ sudo nano /etc/puppetlabs/code/environments/production/manifests/site.pp  
ubuntu@ip-172-31-14-115:~$ sudo nano /etc/puppetlabs/code/environments/production/manifests/site.pp  
ubuntu@ip-172-31-14-115:~$ sudo cat /etc/puppetlabs/code/environments/production/manifests/site.pp  
node 'puppet-agent01'{  
  include centraldemo  
}  
ubuntu@ip-172-31-14-115:~$ sudo cat /etc/puppetlabs/code/environments/production/modules/centraldemo/manifests/init.pp  
class centraldemo{  
  package{'tree':  
    ensure=>installed  
  }  
}  
  
file{'/tmp/master-demo.txt':  
  ensure=>file,  
  content=>"This file is managed by Puppet master.\n",  
  owner=>'root',  
  group=>'root',  
  mode=>'0644'  
}  
  
file{'/opt/company':  
  ensure=>directory,  
  owner=>'root',  
  group=>'root',  
  mode=>'0755'  
}  
}  
ubuntu@ip-172-31-14-115:~$ |
```


Agent:

```
ubuntu@ip-172-31-7-84: ~  
ubuntu@ip-172-31-7-84:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1785778132'  
Notice: /Stage[main]/Centraldemo/Package[tree]/ensure: created  
Notice: /Stage[main]/Centraldemo/File[/tmp/master-demo.txt]/ensure: defined cont  
ent as '[sha256]700e0b50e70e7198848cd408f3ed0a245b3cd11a6f02f66cd3431ec2f28ee63a  
'  
Notice: /Stage[main]/Centraldemo/File[/opt/company]/ensure: created  
Notice: Applied catalog in 4.57 seconds  
ubuntu@ip-172-31-7-84:~$ cat /tmp/master-demo.txt  
This file is managed by Puppet master.  
ubuntu@ip-172-31-7-84:~$ ls -l /opt/company  
total 0  
ubuntu@ip-172-31-7-84:~$ find /opt/company  
/opt/company  
ubuntu@ip-172-31-7-84:~$ ls -ld /opt/company  
drwxr-xr-x 2 root root 4096 Aug  3 17:29 /opt/company  
ubuntu@ip-172-31-7-84:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'
```

```
ubuntu@ip-172-31-7-84: ~  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1785778246'  
Notice: Applied catalog in 0.41 seconds  
ubuntu@ip-172-31-7-84:~$ echo "I hacked this file!" | sudo tee /tmp/master-demo.  
txt  
I hacked this file!  
ubuntu@ip-172-31-7-84:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1785778331'  
Notice: /Stage[main]/Centraldemo/File[/tmp/master-demo.txt]/content:  
--- /tmp/master-demo.txt          2026-08-03 17:32:00.457105606 +0000  
+++ /tmp/puppet-file20260803-2651-iuszho      2026-08-03 17:32:12.750223100 +0  
000  
@@ -1 +1 @@
```

```
ubuntu@ip-172-31-7-84: ~
Info: Retrieving pluginfacts
Info: Retrieving plugin
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-agent01
Info: Applying configuration version '1785778331'
Notice: /Stage[main]/Centraldemo/File[/tmp/master-demo.txt]/content:
--- /tmp/master-demo.txt      2026-08-03 17:32:00.457105606 +0000
+++ /tmp/puppet-file20260803-2651-iuszho      2026-08-03 17:32:12.750223100 +0000
@@ -1 +1 @@
-I hacked this file!
+This file is managed by Puppet master.

Notice: /Stage[main]/Centraldemo/File[/tmp/master-demo.txt]/content:

Notice: /Stage[main]/Centraldemo/File[/tmp/master-demo.txt]/content: content changed '{sha256}55b225424509825bb2128c293d395e6b9d83a67cf81387a2ee62713fdbac6a0e' to '{sha256}700e0b50e70e7198848cd408f3ed0a245b3cd11a6f02f66cd3431ec2f28ee63a' (corrective)
Notice: Applied catalog in 0.43 seconds
ubuntu@ip-172-31-7-84:~$ cat /tmp/master-demo.txt
This file is managed by Puppet master.
ubuntu@ip-172-31-7-84:~$ |
```

Instances used all thru these practicals:

✓ puppet-master	i-0c0b95ffc889341a7	✓ Running	t3.small	⌚ Initializing	ap-south-1b	ec2-3-110-188-5.ap-so...
✓ puppet-agent	i-093cc3c71fb0dd8d8	✓ Running	t3.micro	✓ 3/3 checks passed	ap-south-1b	ec2-13-127-65-166.ap-...

Note: Must say switching from a t3.micro to a t3.small significantly improved responsiveness. Catalog compilation, agent runs, and Puppet Server startup were actually noticeably faster, with no visible lag during execution (the inside details of it is even more interesting if anyone wants to explore)